

Cardmember Profitability & Customer Lifetime Value (CLV) Engine

Executive Case Study | Portfolio Size: 500,000 Accounts | Focus: Premier Charge Card Unit Economics

EXECUTIVE SUMMARY: THE PROBLEM STATEMENT

- Gross spending volume does not directly equate to net issuer profit in ultra-premium card portfolios.
- 'Perk Gamers' exploit 5x travel multipliers and statement credits (\$640+ in airline/cab/entertainment perks) while paying in full, operating at negative gross margins for the issuer.
- True portfolio profitability is driven by Prime Revolvers (generating high-margin Net Interest Income) and loyal, multi-card Transactors whose merchant swipe fees exceed operational perk expenses.

500,000

Total Portfolio Records

11.3x Lift

Revolving Interest Margin

68x Drop

Collection Delinquency Rate

+\$2,276

Mean Top 20% Net Profit

FINANCIAL ARCHITECTURE: CREDIT CARD P&L FORMULATION

Net Profitability = Gross Revenue – Direct Perk Costs – Expected Credit Loss – Delinquency Penalties

1. Revenue Engines (+)

Net Interest Income (NII): $0.20 * f1$

- 20% Net Margin (24% APR - 4% CoF)

Merchant Interchange: $0.027 * f5$

- 2.7% Closed-loop discount rate

Annual Card Fees: $695*f20 + 175*f19$

- $695basecard + 175\text{ supp. cards}$

2. Operational Perk Outflows (-)

Statement Credits: $f14 + 15*f15 + f16$

- \$200 Airline + \$180 Cab + \$240 Digital

Lounge Access Costs: $40 * f13$

- \$40 per Centurion/Priority Pass visit

Rewards Cash Liability: $0.012 * f21$

- 1.2 cents per Membership Reward point

3. Risk & Delinquency (-)

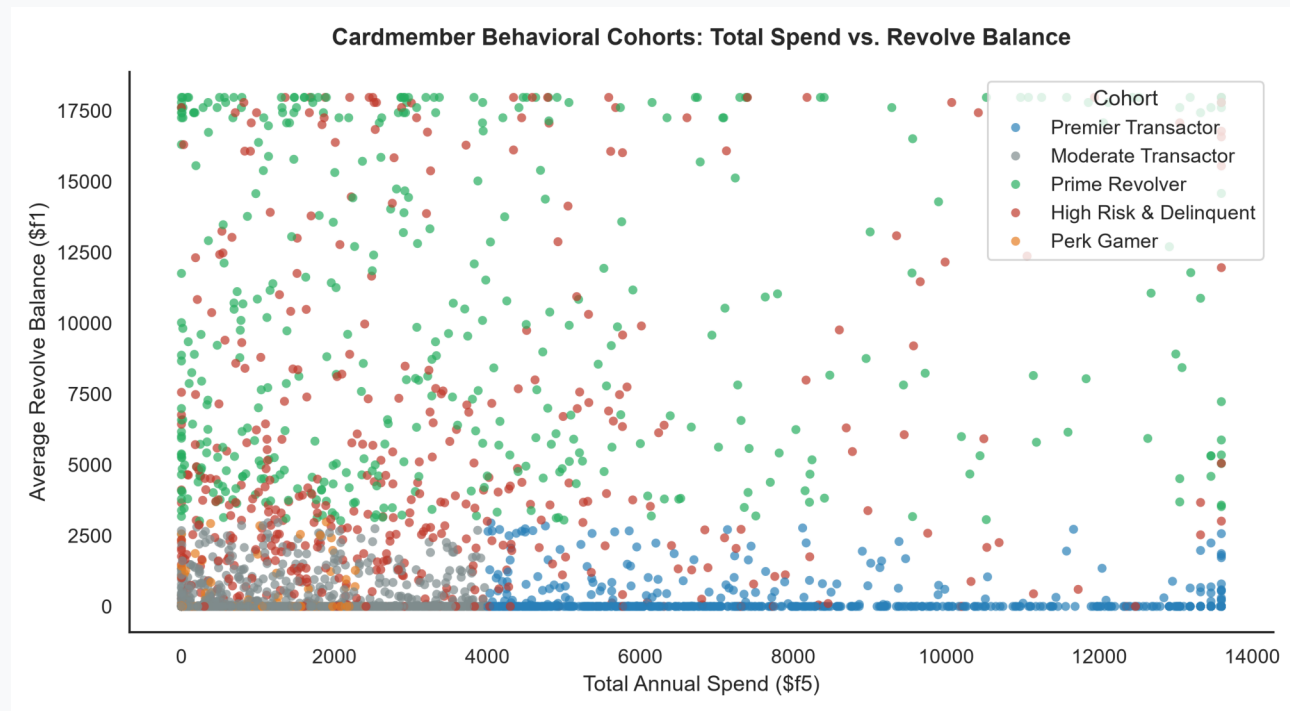
Basel II / IFRS 9 Expected Loss:

- $EL = PD * EAD * LGD$
- $EAD = \max(f17, f1), LGD = 85\%$
- $EL = f11 * \max(f17, f1) * 0.85$

Delinquency Friction: $1500*f3 + 500*f2$

- Severe collection & churn penalties

EXPLORATORY DATA ANALYSIS: BEHAVIORAL COHORTS



PORTFOLIO BEHAVIORAL ARCHETYPES

1. Prime Revolvers (30% Share)

Carries high balances (\$3k+), low default risk.
Primary driver of Net Interest Income (NII).

2. Premier Transactors (45% Share)

High travel spending (\$4k+), multiple cards, pays in full.
Generates high interchange + \$695 annual fee revenue.

3. Perk Gamers (15% Share)

High reward redemptions (>30k pts), claims all perks, low spend.
Generates near-zero or negative net margin.

4. High-Risk / Delinquent (10% Share)

Flagged with collection calls (f3=1) or risk score >0.10.
Accounts routed for immediate risk mitigation.

PORTFOLIO STRATIFICATION: TOP 20% DECILE LIFT



TOP 20% vs. BOTTOM 20% FINANCIAL PROFILE

Metric	Top 20%	Bottom 20%	Impact
Avg Revolve Balance	\$7,637	\$675	11.32x Lift
Annual Spend Volume	\$4,919	\$1,590	3.09x Lift
Airlines & Hotel Spend	\$15,132	\$4,471	3.38x Lift
Collection Call Rate	0.41%	2.68%	6.54x Drop in Bad Debt
Average Risk Score	0.0183	0.0655	72% Lower Risk
Net Contribution (\$)	+\$2,276	+\$6,795	\$4,519 Margin Spread

STRATEGIC ROADMAP & INSTITUTIONAL ASSUMPTIONS

PRESCRIPTIVE BUSINESS ACTION MATRIX

Prime Revolvers (Deciles 9-10)

- Proactive Credit Line Increases (PCLI)
- Promotional Pay-Over-Time APR rates to drive NII

Premier Transactors (Deciles 8-10)

- Bespoke airline/hotel transfer partner bonuses
- Priority concierge retention on \$695 annual renewal

Perk Gamers (Deciles 3-5)

- Dynamic point redemption thresholds
- Restructure multipliers on low-margin categories

Distressed Accounts (Deciles 1-2)

- Immediate credit line freezing and overdraft restriction
- Early loss-mitigation routing before charge-off

KEY BUSINESS ASSUMPTIONS & ALM BENCHMARKS

1. Treasury Cost of Funds (TCF):

Revolving margin (20%) reflects a 24% gross APR minus a 4% Treasury Cost of Funds benchmark.

2. Closed-Loop Network Structure:

Modeled as a proprietary charge card where Amex retains the full net merchant discount fee (~2.7%) without co-brand dilution.

3. Basel II / IFRS 9 Loss Given Default (LGD):

Unsecured revolving recoveries benchmarked at 15%, leaving 85% LGD.

4. Data Warehousing Imputation Logic:

Missing values in left-joined transactional logs represent true \$0 activity; missing risk scores represent portfolio medians.